Andy Welsh

Experienced Cloud and Infrastructure Engineer

andy@tinfoilcipher.co.uk
www.tinfoilcipher.co.uk

LinkedIn: awelshuk
GitHub: tinfoilcipher

EXPERIENCE

Senior Consultant - DevOps, Platform and Cloud, Opencast
June 2022 to Present, Newcastle, United Kingdom

Cloud consultancy services for a number of high-profile public and private sector clients throughout the UK. Providing architecture, deployment and implementation strategy for major cloud platforms (AWS and Azure).

Designing and managing infrastructure, security and deployment strategies up to production for modern container platforms (Kubernetes, Openshift, AWS ECR, Azure Containers) and monolithic cloud-native platforms (EC2 and Azure VMs) as well as integrations between on-premise Linux systems, network infrastructure and a broad range of PaaS and SaaS services.

Providing hands-on technical, security and architectural advice to development teams migrating to container environments. Leading troubleshooting and fault diagnosis for container platforms, surrounding network infrastructure and integrations.

Writing and management of common Terraform modules, common CI/CD templates (in a range of CI and CD tooling) and Ansible/Packer configurations to provide unification between disparate departments.

Management of a number of CI/CD platforms including Gitlab CI/CD, GitHub Actions, Jenkins, Argo CD, FluxCD and Concourse CI.

Rapid development of new proof of concepts for cutting edge applications to test their viability as platform tools.

As a senior engineer the role also involves providing technical advice to junior staff, mentoring and delivering technical talks to internal and client teams.

Senior Platform Engineer, ITV
February 2022 to May 2022, Leeds, United Kingdom

Cloud Platform Engineering for a major UK broadcaster as part of the Online Services team.

Managing several backend components of the common AWS platform serving a community of development teams responsible for nationwide television streaming.

Providing monitoring, support and architectural advice for all teams using the service. Other duties involved performing Terraform and Puppet deployments and upgrades, ensuring that security and deployment patterns are adhered to, configuring and managing CI/CD pipelines (via Jenkins and GitHub Actions), carrying out environment audits and performing code reviews.

DevOps Engineer, Caspian
June 2020 to February 2022, Newcastle, United Kingdom

Cloud DevOps Engineering; management of an industry leading FinTech SaaS platform over multiple public cloud environments (AWS and GCP) in partnership with Nasdaq Tech.

Deployment, configuration and maintenance of Kubernetes clusters (EKS) as well as surrounding AWS PaaS. Management of GCP PaaS and IAM. Configuration and deployment of additional Kubernetes services (Istio Service Mesh, Prometheus, fluentd, flux-cd among others).

Design and implementation of CI/CD strategy and tooling up to production, leveraging Terraform, Helm and Gitlab CI/CD.

Expertise
Infrastructure as Code
Container Orchestration
Public Key Infrastructure
Linux Systems Administration
Infrastructure Security

Server/Cloud
Amazon Web Services
Microsoft Azure
Linux (Ubuntu/RHEL/CentOS)
VMWare ESXi/vCenter
Windows Server
Exchange Server

Platform/Software
Terraform
Kubernetes
Openshift
Istio
Helm
Docker
Crossplane
Ansible/AAP
Packer
Vault
BIND DNS
NGINX
Active Directory

CI/CD
Gitlab CI/CD
GitHub Actions
BitBucket
Concourse CI
Argo CD
FluxCD

Scripting
Bash/Sh
Python
PowerShell

Networking
Juniper JUNOS
Cisco IOS/ASA

Design and implementation of Terraform Modules to allow for highly available, templatable deployment of all SaaS platform environments, as well as all private development and testing environments. Development of testing and release strategy for new modules.

Deployment of centralised management stacks using cloud-native tooling for Secrets Management, IPAM and Helm package management.

Implementation of cloud network security and integrations (Certificate rotation, L7 load balancer deployment, Incapsula WAF configuration, Istio Service Mesh configuration, AWS IAM/S3 policies, AWS KMS).

DevOps Engineer, Fenwick

October 2018 to June 2020, Newcastle, United Kingdom

Cloud/on-premise Infrastructure engineering and modernisation for a nationwide retail estate of Windows Server, Ubuntu/CentOS Linux and Azure platforms.

Design and implementation of Configuration Management and Automation infrastructure for Azure/VMWare estate leveraging Red Hat Ansible Automation Platform, Azure DevOps, and PowerShell/Bash scripting. Design and delivery of source control (git) implementation in to a team previously unfamiliar with source control management.

Design and implementation of estate-wide IPAM and Configuration Management for network infrastructure (leveraging Netbox, NAPALM and RANCID) to provide centralised locations for the store of network configuration and logistical data.

Deployment and administration of Azure Active Directory/IAM and PaaS (SQL, Storage, VM, LogicApps, App Services, KeyVault, Backup, Networks). Maintenance and improvement of Windows, Linux and Cisco centric on-premise and hybrid cloud infrastructure (VMWare ESXi/Azure).

Design and deployment of two-tier PKI implementation to an estate previously leveraging no TLS encryption services.

IT Engineer, RMT Technology

April 2015 to September 2018, Newcastle, United Kingdom

Systems Administration, Architecture and IT support for a Managed Service Provider catering to SMEs. A broad role including, server maintenance and implementation (Windows/Linux), server/desktop support, hardware and comms installation, backup management, patch management, network configuration and troubleshooting, bespoke scripting and application support.

Covering design, implementation and support of primarily Windows Server centric environments for companies of varying sizes within both physical and virtual environments (Hyper-V and VMWare ESXi). Network architecture and management in primarily Juniper and Cisco environments (LAN, WLAN, Client and S2S VPN).

Configuration and support of core Microsoft infrastructure and SaaS technologies (ADDS / Active Directory and Entra ID / Exchange / DNS / DHCP / DFS / SharePoint).

EDUCATION

AWS Certified Cloud Practitioner (CLF-C02)
Hashicorp Certified Terraform Associate (003)
Microsoft Certified Azure Administrator Associate (AZ103)
ISC2 Certified in Cyber Security
ITIL v4 Foundation Certificate in IT Service Management